7
Sovereign Default

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This chapter examines sovereign default. We begin with the challenge of defining default, highlighting the limitations of existing definitions and proposing an alternative. Section 2 considers the sovereign's decision to default, first, by creditor type, second, by the manner of default (debtor action). We illustrate these with case studies of crises in Jamaica, Ukraine, Uruguay, and Russia. Section 3 reviews the economic determinants of default, including domestic and external shocks, and considers the legal determinants, which merit further systematic study. Section 4 surveys the economic, financial, and legal consequences of default. Section 5 concludes with ideas for reducing the incidence and the cost of default. Throughout this chapter, we start from the assumption that default is never the first-best option for the sovereign or its creditors. Nonetheless, we also recognize that default—in the sense of not paying debt in full and on time—is sometimes unavoidable and may be the best choice available to the sovereign debtor under some circumstances. Defaults should not be avoided at all cost, but debtors and creditors alike can help reduce its incidence and cost, consistent with development, growth, and financial stability objectives.

1. Defining Default: Legal and Economic Perspectives

A. The definition challenge: An overview and proposed solution

It is surprisingly hard to define sovereign default. Academic and policy analyses often conflate default—an *event*—with *conditions* that precipitate the event

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(insolvency, illiquidity, unwillingness to pay), with debtor conduct in the debt restructuring *process* (voluntary, coercive), or with restructuring *outcomes* (haircut, maturity extension) and *consequences* (market exclusion).

Default at its simplest is a broken promise. For sovereign debt, it could include a missed payment, involuntary subordination, or data misreporting. In many cases, debt contracts specify violations that trigger particular creditor remedies. The list of violations and remedies is usually found under the heading "Events of Default" (EoD) in English- and New York-law debt contracts and their analogues in official bilateral and multilateral credit agreements. Contractual EoD are designed to be observable and enforceable, whether by giving creditors the right to suspend disbursements and demand immediate repayment in full, or by authorizing judicial remedies.[1]

Formal contractual definitions are both too broad and too narrow to be useful. They are too broad because they include minor infractions, such as a brief delay in transmitting paperwork, which creditors would normally consider unimportant. The definitions are too narrow because they exclude economically significant events such as the Greek debt exchange in March of 2012, in which no payments were missed and no promises were formally broken—contracts were modified instead—but creditors still faced deep losses. The rise in domestic-law debt complicates the analysis further, because such debt often lacks *any* contractual definitions of default, because background law differs across countries, and because sovereigns can change their own laws to undo or excuse default.

We therefore go beyond debt contracts, to review definitions of default used in seminal economic research (Section 1.B) and by market participants, such as credit rating agencies and derivatives traders (Sections 1.F and 1.G). Third-party actions such as ratings downgrades or credit default swap (CDS) triggers can have major economic consequences for the sovereign and its national economy, on par with the consequences of breach. Third-party definitions of default key off discrete, observable events, but tend to focus on economic substance over legal form. We do not cover potentially influential third-party responses to sovereign distress, such as index exclusion, collateral eligibility, or regulatory treatment, which may merit further consideration.

In practice, neither formal contractual nor substantive economic definitions are fully satisfactory. To address some of their shortcomings, we propose an

[1] In contrast, because official creditors do not normally sue sovereign debtors in national courts, the parties might reasonably expect disputes to be resolved through diplomatic or administrative channels.

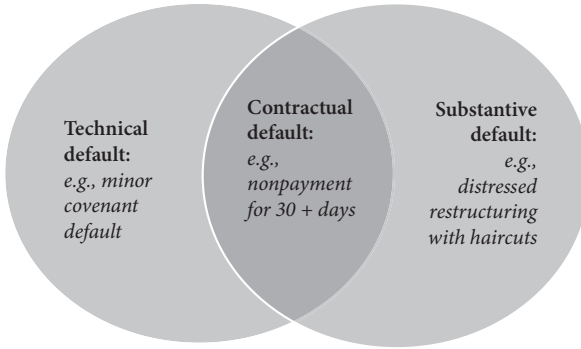

Figure 7.1 Defining default

analytical approach that would distinguish among *technical default*, *contractual default*, and *substantive default*, as follows (also depicted in Figure 7.1):

Technical default includes any contractual EoD or equivalent for domestic and official debt that does *not* also constitute default under specified influential third-party definitions. Administrative errors and some covenant defaults viewed as minor by market participants would fit under this heading.

Contractual default includes the occurrence of any EoD or its functional equivalent that *also* constitutes default under the same third-party definitions. For instance, virtually all contractual and third-party definitions of default include failure to repay principal or pay interest after a grace period (typically 30 days); these would fall under contractual default.

Substantive default includes events that would count as default in third-party documentation and practice, but not in the underlying contract. Thus, a distressed debt exchange, or a restructuring using local law or Collective Action Clauses (CACs), would be substantive default if it complied with the contract and resulted in less favorable terms for the creditor.

We find this approach appealing for its relative simplicity, for its ability to account simultaneously for internal contractual and outside market views of what matters in default, and for disaggregating the consequences of default. The remainder of Section 1 will examine more closely the content, rationale, and operation of theoretical, contractual, and third-party definitions of default, and their implications for research and practice. We further elaborate the prevailing explanations of why sovereigns default and review the relevant literature in Section 2.

B. How economic research defines default

Economic theory literature does not go to great lengths to define the concept of default. Typically, the definition is implicit and includes missed payments of any sort. For example, in Eaton and Gersovitz (1981) a country that misses a single payment will no longer be able to borrow from international capital markets for reputation reasons. As a result, it is optimal for the government to default in full. The definition of default as complete payment cessation lives on in many modern quantitative models building on the Eaton–Gersovitz framework (e.g., Aguiar and Gopinath 2006; Arellano 2008). In this world, default is an all-or-nothing affair.

Bulow and Rogoff (1989a) take a more nuanced approach and allow for partial default and debt renegotiation, with legal considerations playing an important role. If a sovereign misses a payment it breaches its contract with lenders (banks) who can retaliate by seizing part of the country's exports. Moreover, rather than defaulting in full, debtors can negotiate debt restructuring agreements with partial losses. Many recent models follow this approach and also allow for the possibility of debt renegotiations with partial default (e.g. Benjamin and Wright 2009; Yue 2010; or Asonuma and Trebesch 2016). Another strand of the theory literature builds on Grossman and van Huyck (1988), where lenders distinguish between excusable defaults, which occur as a result of exogenous shocks, and unjustifiable repudiation (inexcusable default). While theoretically appealing, inexcusable or strategic defaults are difficult to identify in practice.

In the empirical literature, many economists use rating agencies' definition of default (see Section 1.F below). The most widely used datasets adopt this approach. Reinhart and Rogoff (2009), in particular, trace missed payments as well as distressed debt restructurings to construct a binary indicator of sovereign default. More recently, researchers have refined this approach by calibrating the scope of default events, such as missed payments (arrears), and the size of creditor losses (haircuts) in distressed restructurings in modern and historical debt crises (Sturzenegger and Zettelmeyer 2007; Cruces and Trebesch 2013; Schlegl et al. 2017; Beers and Mavalwalla 2018; Meyer et al. 2019). This is closer in spirit to Bulow and Rogoff (1989a) and moves away from the simplistic zero-one default definition.

Research on the cost of default (Section 4 below) has used many different default definitions in an effort to identify which elements of debt distress and distress management—such as economic conditions, debtor conduct, or restructuring outcomes—have significant economic consequences. However,

using a single term "default" for a variety of events, conditions, and behaviors invites apples-and-oranges comparisons by casual or opportunistic consumers of the studies.

C. Contractual events of default: market instruments

The following categories of contractual EoD are typical in the sovereign debt markets in London and New York:

Payment default is failure to pay principal, interest, or other amounts (such as tax gross-up) when due, after the expiration of any applicable grace period. Interest payments typically enjoy grace periods ranging from ten to thirty days. Grace periods are slightly less common, and may be shorter, for principal payments (e.g., Gooch and Klein 1994; Buchheit 2006 for loans; *see* Annex 7.1 for examples from tradable securities).

Payment, settlement, and securities custody mechanics can make it hard to ascertain the precise timing of payment default. Debtors and creditors typically deal with each other through layers of agents. Contracts typically say that payment is made when the debtor has transferred funds to the paying agent, trustee, or clearing system. However, some contracts do not consider a payment to be made until each creditor has received the funds, as in Argentina's 2005 and 2010 exchange bonds. This distinction became salient when a US federal court blocked Bank of New York Mellon as trustee for Argentina's exchange bonds from distributing the government's interest payment to the bondholders.

Repudiation happens when the sovereign rejects its obligation to pay, which could happen before or after any payment is due. Governments typically avoid questioning the validity of their debts, or announcing their intention not to pay before missing a payment, since in practice, repudiation carries all the traumatic consequences of payment default discussed in Section 4. Repudiation may go hand in hand with governments questioning the legitimacy of one or more obligations. The literature on "Odious Debt" includes a handful of examples (Lienau 2014; King 2016). In 2008, Ecuador claimed that two bonds issued by a previous government were illegitimate and pledged not to pay them. It launched a buyback offer the following year in the shadow of the illegitimacy claim, and ultimately secured nearly two-thirds debt relief with more than 90 percent of the creditors participating.

Moratorium is a unilateral payment stop on one or more debt obligations. The sovereign might announce a moratorium—as Mexico did in 1982

(Kraft 1984)—as an interim measure before launching a debt restructuring; it might also stop payments indefinitely. A moratorium entails a public act, such as an announcement or legislation, apart from the missed payment, which can come before or after the payment default. However, it need not contest the validity of the underlying obligation. A moratorium is distinct from a negotiated payment suspension: if the creditors agree to a stop, they can waive the payment default.

Policy-related EoD may include loss of IMF membership or ineligibility to draw on IMF resources. Such EoD have the practical consequence of incorporating elements of the IMF Articles of Agreement and policies, and amplify the effect of its sanctions (e.g., Choi et al. 2012). Other policy-related EoD, such as maximum debt ratios, are widespread in corporate debt but unusual among sovereigns. Notable exceptions include Ukraine's borrowing from Russia in 2014, which included a number of bespoke EoD designed to maximize creditor control and make it easy to trigger acceleration. Because policy conditions are often at the heart of official lending, some forms of multilateral policy conditionality are indirectly incorporated in contractual EoD by reference to membership and sanctions.

Covenant default is a residual category that captures a breach of all other express promises under the debt agreement (Annex 7.1), ranging from clerical omissions to material violations. The latter category might include effective subordination of creditors without their consent, in violation of the *pari passu* or negative pledge clauses. Covenant defaults also include false representations, which could range from data misreporting to lack of borrowing authority at the time the contract was made. Subsequent loss of borrowing authority is usually a separate EoD.

Cross-default terms link two otherwise-unrelated debt contracts, so that default under one becomes default under the other (Annex 7.1). The theory behind cross-default is a mix of early warning and inter-creditor equity. All else being equal, missing payments to other creditors points to debt distress. Without cross-default, a creditor may have no recourse as others sue and divide up the debtor's scarce assets among themselves.

Cross-default clauses vary in two important ways: trigger and scope. With hair-trigger cross-default, creditors may exercise their default remedies in response to a minor infraction under someone else's contract. At the other extreme, they might have to wait until creditors under the other contract have demanded immediate repayment in full. Some versions of the clause excerpted in Annex 1 include minimum thresholds for missed payments on other debts. The scope of cross-default can range from a narrow sliver of

similar debt (e.g., foreign-law bonds cross-defaulting to foreign-law bonds) to all sovereign and quasi-sovereign obligations, which is rare.

Four additional observations should help situate contractual EoD in the sovereign default context:

First, EoD only give creditors the right to invoke contractual remedies; creditors are under no obligation to do so. In prominent cases of selective default (see Section 2.B), including most recently Venezuela, bond holders chose not to exercise their rights long after the governments defaulted on other debt, preferring instead to get paid as long as possible. If creditors do not act, the debtor may have an opportunity to "cure" the default.

Second, a debt restructuring may not constitute a contractual EoD, regardless of creditor losses. As noted earlier, a market-based debt exchange, a voluntary renegotiation, or a majority vote to change debt terms using CACs would either follow the contract or circumvent it. Neither would breach it.

Third, EoD in sovereign bond contracts are often subject to collective action requirements. For instance, even if the government fails to make a scheduled interest payment and the grace period expires, bond holders may have to muster a vote of at least 25 percent of the principal to instruct the Trustee to "accelerate." If the debtor later makes up the payment, holders of at least 50 percent of the principal could instruct the Trustee to reverse the acceleration (Buchheit and Gulati 2002).

Fourth, minor differences in EoD wording and procedural requirements, such as notices, can lead to different consequences for the same events.

D. Contractual defaults on official debt—suspension, refund, acceleration

Official bilateral and multilateral credits include terms similar to private contractual EoD, but are structured differently in important ways, reflecting the creditors' mandates and, for multilateral creditors, their character as membership organizations. We use the General Conditions of IBRD and IDA (together, World Bank) loans to illustrate (World Bank Group 2017). The General Conditions make a useful point of comparison to private contractual EoD because the World Bank is the largest multilateral creditor, it has a global policy mandate, it lends only to sovereigns, and because today's General Conditions have been revised many times to reflect the World Bank's experience in lending.

The General Conditions are incorporated by reference in transaction-specific "legal agreements" between the World Bank and its borrowing member; unlike the legal agreements, General Conditions do not vary by borrower or by transaction. Article VII of General Conditions contains three potential analogues to contractual EoD: (i) events that would allow the World Bank to suspend or cancel disbursements; (ii) events that would require a refund from the sovereign; and (iii) events that could trigger acceleration (immediate repayment). The last category, "Events of Acceleration," comes closest to EoD in private contracts, and can trigger cross-default under private contracts.

Suspension and cancellation

The World Bank may suspend disbursements or cancel the loan in response to any of the following: "payment failure," "performance failure" (referencing transaction-specific terms), fraud, corruption, misrepresentation, unauthorized assignment of obligations, ineligibility to draw, withdrawal from membership in the World Bank or the IMF, "cross-suspension" of other World Bank loans, or any of a series of events that convince the lender that the program is unlikely to be carried out. Suspension and cancellation events are distinct from default under market instruments. The focus is on the policy objectives and the use of proceeds consistent with the lender's mandate. Ideally, the prospect of suspension of all World Bank disbursements across the board should bring the authorities to the negotiating table and fix the underlying problem. However, it is not meant to bring about the collapse of the sovereign's debt structure through cross-default.

Refund

The World Bank can require sovereigns to refund past disbursements if it determines that they were used in a manner inconsistent with its loan agreement, typically due to fraud or corruption. The 2017 revision made clear that the refund requirement was not intended, and was not perceived in the market, as a cross-default trigger for bonds and loans.

Acceleration

The World Bank can demand immediate repayment of a disbursed loan in the event of a payment default on any of its exposure to the sovereign, subject to a grace period. Unauthorized assignment, material adverse change, failure of co-financing, and events specified in transaction-specific agreements may trigger acceleration under some circumstances. Of all the sanction triggers

specified in Article VII of General Conditions, the term "default" is only used in the third category, "Events of Acceleration," which is understood to interact with cross-default terms in loan and bond contracts.

E. Domestic debt—no definition, no default?

Debt governed by the sovereign issuer's domestic law typically has few express terms. For example, it is not customary for domestic-law debt to include a litany of EoD. As a result, it can be difficult to identify a clear contractual definition of default on domestic debt. The relevant contractual terms may be incorporated by reference to statutes and administrative regulations, which vary in form and substance among different countries (Addo Awadzi 2015). For instance, the Uniform Offering Circular is a US federal regulation that sets out terms and conditions for most tradable US Treasury securities. Of its thirty-four sections, thirty spell out auction procedures; one tells when the creditors are paid, and none address substantive modification or default (31 C.F.R. 356 and 356.30).

This does not mean that governments cannot default on domestic debt. Instead, default and remedies for default are a matter of background law—contract, constitutional, and administrative, among others. For instance, in most jurisdictions, a debtor that simply fails to pay as promised would be in default. However, as Austin (2016) illustrates with examples of payment disruptions on US Treasury securities in 1814, 1933, and 1979, missing domestic debt payments sometimes has no discernible domestic or international consequences for the sovereign. He suggests that of the three incidents, the failure to pay in 1814 on account of a "bankrupt" Treasury comes closest to the widely shared contemporary understanding of default. There is no agreement in the literature or jurisprudence on the default status of the other two episodes, which involved retroactive removal of indexation and administrative delays, respectively.

The power to change domestic law—so that default is either excused, or no longer counts as default—is inherent in sovereignty, subject only to constitutional constraints as interpreted and enforced by domestic courts. In some legal systems, governments have express additional flexibility under domestic law to respond to economic emergencies (e.g., Gross and Ni Aiolan 2006).[2]

[2] For a recent illustration, see e.g., *Mamatas and Others v. Greece*, ECHR 256 (2016), 21.07.2016.

In sum, even something as simple as payment default on domestic debt requires context for a meaningful definition. It is not clear that an event with no discernible legal or economic consequences should be called a default.

Although domestic-law sovereign debt may be less vulnerable to formal default, it is more vulnerable to unilateral modification by the debtor designed to lower its payment burden and reduce payoff for the creditors (e.g., Moody's 2011; Cruces and Trebesch 2013; S&P 2017; Beers and Mavalwalla 2018; Fitch 2018). If debt is denominated in local currency, sovereigns can use monetary policy to reduce its value (Chapter 6). They can also use fiscal policy, such as withholding taxes, to recapture at least in part payments that would otherwise go to creditors.[3] Reinhart and Sbrancia (2015) group these policies under the rubric of modern-day "financial repression."

F. Rating agency criteria for default

Rating agency definitions of default matter because they inform ratings actions. A sovereign downgrade may lead to rapid sell-off of the borrower's debt, since some investors would be barred from holding it by regulation, contract, or mandate (Böninghausen and Zabel 2015). It may also trigger downgrades of other borrowers in the country, particularly financial institutions that benefit from sovereign guarantees, becoming a source of contagion.

As information intermediaries, credit rating agencies have developed distinct methodologies for evaluating sovereign default, which inform their analysis and ratings. Their criteria focus overwhelmingly on payoff. They reference underlying credit agreements but are both far more streamlined and broader than EoD. For example, Moody's definition of default includes three kinds of events:

(i) failure to pay interest or principal within the grace period under the debt agreement;
(ii) a distressed debt exchange that reduces the sovereign's financial obligation to avoid payment default; and
(iii) unilateral change in payment terms "imposed by the sovereign that results in a diminished financial obligation, such as a forced currency

[3] For example, in 1999, the government of Turkey imposed a retroactive withholding tax of between 4 and 19 percent on interest income from domestic currency bonds. Note, however, that many bond contracts include explicit language fixing bondholders' tax liability; imposition of a tax in such a case could be a contractual EoD.

> re-denomination . . . or a forced change in some other aspect of the original promise, such as indexation or maturity."[4] (Moody's 2019)

This definition relies on the underlying agreements for payment default parameters and includes two additional elements beyond payment default: a debt restructuring, even if it is preemptive and consensual (discussed in Chapter 8), and domestic law measures that target and adversely affect payoff, whether or not they amount to default under the terms of domestic debt instruments.

G. Credit default swaps—credit events and default

Sovereign CDS are tradable contracts under which "protection sellers" take on sovereign credit risk for a fee from "protection buyers." The buyers enter into CDS contracts either to hedge existing exposure to the sovereign, or to bet against the sovereign credit. If the sovereign defaults—in CDS parlance, if a "credit event" occurs—the seller must compensate the buyer for the loss in value of a "reference obligation" specified in the CDS contract. Sovereign CDS have grown as a share of the CDS market (primarily attributable to contracts referencing European sovereign credits), reaching 16 percent at the end of 2017. However, the aggregate notional amount of all sovereign CDS outstanding, approximately $1.5 trillion, is still small relative to the $40 trillion bond market (Aldasoro and Ehlers 2018).

CDS definitions of sovereign credit events matter because CDS contracts transmit credit risk and can become a source of financial market contagion in a sovereign debt crisis. CDS contracts are highly standardized and, to enhance CDS liquidity, definitions do not normally vary across parties or transactions. They are drafted by a trade group, the International Swaps and Derivatives Association (ISDA), which takes copyright in its standard terms. The term "Credit Event" is included in the ISDA's Definitions Booklet (ISDA 2003, 2014) and incorporated by reference in particular transactions. CDS could trigger independently of any definition of default in a sovereign debt contract. As a result, ISDA definitions can have a homogenizing effect in a world of incompletely standardized debt contracts.

The definition of sovereign credit events that trigger protection sellers' payment obligations reflect the objectives of the CDS instrument: to isolate and

[4] A fourth event of default, bankruptcy, or receivership, does not apply to sovereigns.

transfer credit risk, as distinct from "legal" risks, economic conditions, or policy performance. In addition, CDS are meant to be actively traded, which implies that credit event attributes should be observable and verifiable.

Among credit events, the definition of restructuring is the most challenging and controversial of the lot, since it captures a wide range of consensual and involuntary outcomes (e.g., Gelpern and Gulati 2012). It includes principal and interest reductions, payment date extensions, subordination, and redenomination into currencies other than those of the G-7 or top-rated OECD member countries, *provided* any such change is related to deteriorating creditworthiness or financial condition of the sovereign and is effected in a way that "binds all holders" of the reference obligation. Protection sellers and buyers can select which of the credit events would apply to their transaction.

2. Varieties of Default: Who is Affected? How is It Done?

While it is never the first-best solution, default can be a critical step on the sovereign's path to debt relief and debt sustainability. The threat of default is among the most powerful tools in the sovereign debtor's restructuring repertoire (see Chapter 8). As noted in Section 1.B, an influential early strand of economic literature has treated debt default as binary—the country is either in default or not (e.g., Eaton and Gersovitz 1981; Arellano 2008). More recent work has begun to delve more deeply into the different ways a sovereign can default. In order to explore the different considerations a debtor must weigh, we examine this decision from two angles: default by creditor type (i.e., on whom to default) and default by debtor action (i.e., how to default).

A. Default by creditor type

Given that a default on different creditor groups will result in different consequences for a debtor, a debtor may choose to discriminate, often by using different categories of debt as a proxy. One key underlying concern in differentiating among creditors is that selective default may give rise to inter-creditor equity concerns and complicate the restructuring task down the road if the debtor's actions are widely perceived as unfair. Whenever a debtor chooses to differentiate, therefore, it becomes important to justify that choice on relevant grounds.

Default on official and private creditors

Among multilateral, bilateral, and private creditors, multilateral creditors are least likely to face sovereign default (Schlegl et al. 2017), and even less likely to participate in a restructuring.[5] It has been generally accepted by official and private creditors that IMF financing, in particular, should be excluded from sovereign debt restructurings, as the IMF's lending during crisis situations (just when all other creditors are exiting) constitutes a public good that helps resolve a country's balance-of-payments problems (Rieffel 2003; IMF 2009; Lastra 2014; Steinkamp and Westermann 2014).[6] Other multilaterals are also generally considered senior creditors (*cf.* Roubini and Setser 2004), though that status has occasionally been called into question (Bulow and Rogoff 1988, Gelpern 2004). Bulow and Rogoff (1988) observe that the data from the 1980s do "not square with the official view that obligations to the IMF and the World Bank are senior claims."

By contrast, official bilateral creditors restructure frequently—pre- or post-default, formally and informally—either through the provision of new financing or the restructuring of existing debt. Indeed, because of the long-standing track record of official creditors giving concessional treatment to distressed sovereign debtors (see Chapter 8), credit rating agencies generally do not consider a failure to pay debt owed to another government a default (e.g., S&P Global Ratings 2017; Fitch 2018).

Conventional wisdom has been that, despite the lack of de jure seniority rankings among creditors, private creditors generally face a higher risk of default and steeper haircuts than official bilateral creditors (e.g., Roubini and Setser 2004).[7] In line with this, Steinkamp and Westermann (2014) note that 65 percent of experts responding to the 2013 World Economic Survey indicated that they expected bilateral loans extended during the Eurozone crisis to be treated as senior debt. However, recent research challenges whether this

[5] For example, Reinhart and Trebesch (2015) show twenty-three instances of members running arrears to the IMF over the institution's 70-year history.

At the start of the twenty-first century, international pressure prompted some of the largest multilateral creditors, including the World Bank and the IMF, to provide conditional debt relief to a group of low-income countries through the Heavily Indebted Poor Countries Initiative and the Multilateral Debt Relief Initiative, with assurances from the G-8 countries that such debt relief would jeopardize neither the multilaterals' ability to continue to provide financial support nor the multilaterals' own finances.

[6] The de facto nature of the IMF's "preferred creditor status" means that when a country receives financing from the IMF, there is no legal subordination of existing debt, and no credit event has occurred. For an example, see the International Swaps and Derivative Association's determination regarding Ireland's IMF financing in 2011.

[7] Trade creditors are often seen as outside this calculus because interruption of payments would have immediate implications for trade (Kaletsky 1985). More recent work, however, has found that trade creditors face default more often than previously supposed (Schlegl et al. 2017).

perceived seniority holds true in practice. Schlegl et al. (2017) and Moody's (2018) both find that Paris Club restructurings outnumber defaults on private creditors and often result in larger haircuts on the official creditors.[8] Moreover, when focusing on the start of default, Schlegl et al. (2017) find that sovereigns are more likely to accumulate arrears towards official creditors than towards private creditors. The speed and scope at which payments are missed suggests that government-to-government loans are junior to private bank loans and bonds, even after controlling for country characteristics and the size and composition of the debt outstanding.

Default on foreign and domestic creditors

A debtor may also seek to differentiate between foreign and domestic creditors and favor one or the other depending on their domestic concerns and their objectives in the ultimate restructuring (Sturzenegger and Zettelmeyer 2005). Erce and Díaz-Cassou (2010) have found that considerations that lead to this type of discrimination include the origin of liquidity pressures, the soundness of the banking system, and the domestic private sector's reliance on international markets. Others have identified domestic politics as a key factor (Kohlscheen 2010; Erce 2013). The economic consequences of default will be described in more detail below. In short, a default on resident creditors can impact the health of the financial system and will merely reallocate adjustment internally, whereas a default on non-resident creditors can impair private-sector access to capital markets but also will redistribute the burden partially outside the issuer's economy (Erce and Díaz-Cassou 2010; Erce 2013).

Domestic creditors can be subject to certain incentives to participate in an exchange. While a distressed exchange can occur with both domestic and foreign debt, domestic creditors may be particularly vulnerable to the issuer's regulatory power over financial institutions and moral suasion (e.g., appeals to patriotism to increase exposure to government debt).[9] For example, in the 2003 Uruguay restructuring, the central bank declared the old bonds ineligible for liquidity assistance, effectively rendering them unmarketable. Failure of a bank to participate in the exchange would have therefore hurt their provisioning and capital adequacy ratios (IMF 2014). Where the stability of the financial sector was a concern, some restructurings have included a framework for central bank liquidity provision—Case Study 7.1 discusses the case

[8] As highlighted by Roubini and Setser (2004), the differences between Paris Club treatments and private creditor restructurings—e.g., flow treatments by the Paris Club over a limited period compared with "stock" restructurings of privately held debt—makes an apples-to-apples comparison difficult.

[9] Erce and Díaz-Cassou (2010) provide examples from Uruguay's and Argentina's exchanges.

of Jamaica, and Case Study 7.2 presents Uruguay. Russia provides an example of a sovereign default that had devastating effects on the domestic banks (Case Study 7.2).

In practice, cleanly separating foreign and domestic creditors is very difficult. The type of debt can be used as a proxy to some extent, categorizing along different axes, including governing law and currency (Chapter 2). However, as many observers have noted, the move toward liberalizing capital flows in recent years means that foreign creditors are increasingly holding domestic-law, domestic-currency instruments, and vice versa, making this distinction—and the resulting economic predictions—ever more difficult (Gelpern and Setser 2004; IMF 2015; Moody's 2016).

Jamaica's restructuring in 2013 and Ukraine's restructuring in 2015 provide an interesting contrast between debtors that chose to focus on debt held by domestic creditors and foreign creditors, respectively (See Case Study 7.1).

Case Study 7.1 Restructuring domestic vs. external debt

Jamaica (2010, 2013)

In the decade leading up to the 2010 debt exchange, Jamaica faced annual debt service costs of 112 percent of government revenue, with interest on some local currency bonds reaching 28 percent. By 2009, annual interest payments constituted over 60 percent of fiscal revenue.

In January 2010, with a debt-to-GDP ratio of 124 percent, the government launched a pre-default debt exchange chiefly aimed at reducing the fiscal burden of domestic debt service; external debt was explicitly excluded. The 2010 debt exchange (also known as the Jamaica Debt Exchange, or JDX) covered over 350 domestic debt instruments (local currency, USD-indexed, and USD-denominated)—constituting USD 7.86 billion, or 63.7 percent of GDP. Following informal creditor consultations, the exchange involved a reduction of coupons and an extension of maturities with an NPV haircut of approximately 20 percent. The exchange, which was intentionally designed to be light so as to avoid domestic disruption, was completed just one month later in February 2010 with over 99 percent participation and resulted in fiscal savings of 3.5 percent of GDP, an average extension of maturities for domestic debt from 4.7 years to 8.3 years, and an average coupon decline of 650 bps to 12.5 percent. There was no official-sector restructuring.

Continued

Case Study 7.1 *Continued*

Jamaica's credit rating was initially downgraded to Selected Default until the IMF approved a stand-by arrangement for Jamaica. Jamaica reentered the domestic capital market in April 2010 at lower rates than before the JDX, and it raised funds in the international capital market in February 2011 on an oversubscribed bond issuance.

In the end, however, the exchange provided only a temporary reprieve. The following years saw slow growth, continued high government spending, and weak tax compliance. By the end of 2012, the debt-to-GDP ratio stood at about 150 percent. In February 2013, the government announced another preemptive restructuring of domestic debt (this time known as the National Debt Exchange, or NDX). The restructuring, which again reduced coupons and extended maturities, included twenty-eight domestically-held local-currency and USD-denominated bonds amounting to approximately USD 9.1 billion, or 64 percent of GDP. With participation of nearly 100 percent, the restructuring was designed to achieve fiscal savings of 8.5 percent of GDP and involved an NPV haircut of about 10 percent with maturities extended by three to ten years, and coupons lowered between 75 and 500 bps. Rating agencies downgraded Jamaica's credit rating to Selected Default, raising it after the exchange, but not to pre-NDX levels (CCC vs. B–).

Because 65 percent of government debt was held by domestic financial institutions at the time of both restructurings, the government proceeded with caution to avoid threats to domestic financial stability. The government performed stress tests to identify vulnerabilities in the financial system and tailored the exchange accordingly. Jamaica explicitly excluded foreign-law bonds from the restructurings for several reasons. First, domestic debt presented the largest ongoing payment concern. Second, the government wished to quickly reestablish access to international markets. Third, the government lacked sufficient information about foreign bondholders to help secure adequate participation.

Sources: IMF (2014, 2015a, at Annex II, Appendix); Grigorian et al. (2012); Erce (2013); IMF (2013); IMF (2018, Annex II).

Ukraine (2015)

Heightened tensions with Russia following the annexation of Crimea exacerbated ballooning financing needs, and it became clear by end-2014 that Ukraine's debt burden was unsustainable. A preemptive debt exchange operation was announced in January 2015 and launched in September 2015,

with three objectives, tied to the parameters of an IMF-supported program: (i) generating US$15 billion in public sector financing over the next three years; (ii) lowering debt-to-GDP ratio from almost 80 percent to under 71 percent by 2020; and (iii) limiting gross financing needs to an average of 12 percent of GDP in 2015–18 and 10 percent of GDP in 2019–25.

The debt exchange covered four categories of debt—US$18 billion of government-issued Eurobonds held by external creditors including Russia, US$0.5 million of government-guaranteed external commercial loans of SOEs, US$0.6 million of City of Kyiv Eurobonds, and US$3.3 billion of non-sovereign-guaranteed external debt of three SOEs. Following extensive discussions with a creditor committee representing large bondholders, the restructuring called for a 20 percent haircut on outstanding Eurobond amounts, a four-year extension of maturities, and a higher coupon (7.75 vs. 7.2). The exchange also included a GDP warrant to provide a potential upside to creditors in 2021–40. The terms for guaranteed SOE loans and City of Kyiv Eurobonds mirrored those for the Eurobonds but with a 25 percent haircut and shorter maturity extension. There was no haircut for non-guaranteed SOE debts, which were stretched out with a higher coupon. CACs were triggered on the two sets of Eurobonds and the non-guaranteed SOE debt, resulting in no holdouts save Russia's National Wealth Fund (see next paragraph). Participation in the other debt categories was lower.

While debt held by official creditors was generally not restructured, a US$3 billion Eurobond held by Russia's National Wealth Fund was initially swept into the Eurobond restructuring by Ukraine. Russia refused to participate, and Ukraine defaulted on the bond in December 2015. Upon petition by Russia, the IMF declared the Russian-held Eurobond to constitute official bilateral debt under the IMF's arrears policies (see Chapter 8), prompting Ukraine to enter into bilateral discussions on restructuring. Russia filed suit in an English court to enforce payment. At the time of this publication, the debt had not been restructured.

Two-thirds of public debt was held by non-residents; domestic creditors, including those holding foreign-currency debt, were excluded. This perimeter was drawn partly due to financial stability concerns but primarily because increased recapitalization needs would incur a fiscal cost, leading to non-observance of the objectives laid out under the IMF-supported program.

Sources: IMF (2015b); Ministry of Finance of Ukraine (2015); IMF (2016a, Debt Sustainability Analysis Annex); IMF (2016b).

Case Study 7.2 Hard vs, soft restructuring

Uruguay (2003)

Uruguay's economy was severely impacted by the Brazilian and Argentine crises of the late 1990s and early 2000s, in part due to high dollarization. In May 2002, widespread withdrawals from the banking system, including by Argentine depositors, who held 40 percent of deposits in Uruguayan banks, led the Uruguayan authorities to declare a five-day bank holiday. Facing low foreign exchange reserves and crippling external debt-service payments, the government decided to float the currency. The peso depreciated by 27 percent overnight and ultimately by 50 percent. Public debt, which constituted 40 percent of GDP in 2001, reached approximately 100 percent in 2003.

Despite an investment-grade rating as late as 2002, in March 2003, the government announced a preemptive and voluntary debt exchange involving an extension of maturities. The exchange covered US$5.4 billion in foreign-currency debt held by private creditors, including domestically-issued bonds and bills, a Samurai bond issued in Japan, and international bonds issued under foreign law. This was about half of total debt and was considered comprehensive. Domestic creditors were thought to hold just over half of debt and to mostly constitute retail investors. Official bilateral debt, which was not a major component of debt at the time, was not rescheduled. The exchange was launched in April and settled in May with a small NPV reduction. Maturities were extended by an average of 6.4 years for foreign-law bonds and 8.6 years for domestically-issued bonds.

The exchange achieved a high rate of participation—99 percent participation in the domestic exchange, and 93 percent overall—through a combination of techniques (see Chapter 8). Only the relatively small (US$0.3 billion) Samurai bond included a CAC. For the other bonds, participating creditors could choose whether to vote for exit consents to remove the cross-default/cross-acceleration clause, to allow the bonds to be delisted in international exchanges, and to amend the waiver of sovereign immunity. For domestic bank bondholders, the central bank encouraged participation in the exchange by announcing that old bonds would no longer be eligible for liquidity assistance. Because the old bonds had to be marked to market and would require a 100 percent risk weight for capital ratios, it was an inevitability that banks would participate in the exchange. To minimize financial instability, the government, supported by multilateral institutions, established a bank recapitalization facility. This was designed to support the central bank's lender-of-last resort facilities, to provide financing for

bank recapitalization, to cover all US dollar deposits in public banks, and to provide liquidity support to the payment system. Ultimately, the FDBS helped restore confidence in the financial system.

Following the exchange, and supported by strong economic policies and an IMF arrangement, Uruguay reestablished debt sustainability and entered a period of economic recovery. The country regained market access quickly, issuing a dollar-denominated bond in June 2003. Uruguay's credit rating, which was downgraded to "Selective Default" for less than a month, recovered over the following year but did not reach investment grade until 2012.

Sources: Buchheit and Gulati (2000); IMF (2003a, 2003b); Buchheit and Pam (2004); Erce (2013); Cruces and Trebesch (2013); IMF (2014, Annexes II, III, IV).

Russia (1998)

Oil price shocks and the Asian Financial Crisis exacerbated domestic political tensions and financial market pressures in early 1998. By May, interest rates quintupled, and the central bank doubled its sales of foreign exchange to defend the ruble. The government's debt service on short-term government bonds (GKOs) skyrocketed, but investor flight continued, and international reserves plummeted. In mid-July, supported by the IMF, the government announced a voluntary swap of US$4.4 billion of GKOs for long-term Eurobonds. However, very low participation in the exchange, coupled with the political failure of key fiscal measures intended to support the exchange, triggered a rise in GKO yields to nearly 300 percent. The central bank faced pressures from multiple sides—providing credit to the government, supporting commercial banks, and draining reserves to support the ruble. Between July 10 and August 14, the central bank lost US$4.5 billion in reserves.

In August 1998, the government suspended payments on US$45 billion (at the pre-crisis exchange rate) in treasury bills (GKOs and medium-term ruble bonds known as OFZs) maturing before end-1999 and announced a 90-day standstill in servicing private external debts, short positions on currency forwards, and margin calls on repo operations. Because domestic debt represented the biggest liquidity constraint, the government was able to remain current on post-Soviet external debt obligations.[1] Also in August, the government widened the exchange rate band, suspended secondary market trading of GKOs and OFZs, and announced plans to introduce capital controls. Despite large-scale interventions, the central bank was unable to maintain the band, and the ruble was allowed to float in early

[1] The government defaulted on external Soviet-era debt.

Continued

Case Study 7.2 *Continued*

September. The float, coupled with the default on GKOs and OFZs, led to the collapse of many domestic banks, which had invested heavily in government securities. The banking system broke down, with interbank transactions collapsing and the payments system paralyzed.

The debt restructuring was ultimately concluded two years following the default and encompassed US$71.6 billion at pre-crisis exchange rate levels and involved private sector restructuring, a Paris Club treatment, and a London Club restructuring. The exchange with private creditors took only six months. Though the offer was criticized as unilaterally imposed and discriminatory, it received nearly 99 percent support. Non-residents ultimately received approximately five cents on the dollar, though the loss was largely attributable to devaluation. The Paris Club agreement, reached in January 1999, provided a flow rescheduling for Soviet-era debt of US$8 billion—or 4.6 percent of public debt. Agreement was reached with the London Club[2] in August 2000 and provided an over 50 percent haircut on $31.9 billion in bank loans.

Russia recovered from the crisis over the following years and was again rated as investment grade by Moody's in October 2003.

[2] While initially led by the London Club Bank Advisory Committee, the committee of nineteen international banks broke down, and creditors ultimately exchanged their debt bilaterally.

Sources: IMF (1999a, 1999b, pp. 107–12); Kharas et al. (2001); Chiodo and Owyang (2002); Sturzenegger Zettelmeyer (2005); Olivares-Caminal (2009, ch. 4); Pinto and Ulatov (2010); Erce (2013).

Default on banks and bondholders

Rieffel (2003) and others have suggested that "there is a general impression that bonds are senior to bank loans." This observation has empirical support, with Schlegl et al. (2017) finding bank loans more likely to face payment arrears than bonds, especially in recent years.[10] Over the past forty years, it is also true that bank loans of emerging market sovereigns have been restructured more frequently than bonds (Cruces and Trebesch 2013); it remains to be seen if the trend continues in the face of the increasing share of sovereign bonded debt.

[10] Bonds appear to have become more senior over time. The data show that during the 1980s and 1990s, the relative scale of arrears on bonds and bank loans are comparable. Indeed, one of the reasons for the boom in bank lending to sovereigns in the 1970s was a belief that bank loans were *harder* to renegotiate than bonds.

It is important to note that a default on bonds does not limit the damage to one type of creditor; bondholders include investment funds, pension funds, and even official entities such as central banks and sovereign wealth funds. Importantly, banks themselves are fairly large holders of government bonds, making the distinction between bank loans and bonds artificial when looking at the effect on banks. For example, Gennaioli et al. (2014) find that default on bonds can decrease the liquidity of domestic banks, particularly in countries with better-developed financial institutions (see Section 4 below on the cost of default).

B. Default by debtor action

Defaults can also be categorized by actions that the debtor takes, irrespective of which creditors stand on the other side.

Technical default

"Technical default" is not a legal term. As we suggest in the first part of this chapter, the phrase connotes a formal but ultimately unimportant breach. Which breach is important is in the eye of the beholder (hence our proposed choice of reputable and impactful third-party definitions). For example, the European Banking Authority (2017) defines a technical default (also called a "technical past due situation") as occurring only where the default was the result of (a) a "data or system error of the institution"; (b) "failure of the payment system"; (c) the time lag between payment and receipt; or (d) certain specific issues in factoring arrangements. Some others consider a "technical default" a non-payment that is cured within three months without an announcement of default (e.g., Schwarcz 2014).[11]

A few examples of historical defaults that some observers consider "technical" demonstrate to what type of events the term generally applies. In July 1998, Venezuela missed a payment on a local-currency bond with no grace period. The coupons were paid with a one-week delay, and the government claimed the problem was that the check signatory had been unavailable. The credit ratings agency Moody's considered this a technical default but, due to a pattern of similar missed payments, downgraded Venezuela's rating from

[11] Some have also defined "technical default" in a broader sense, including episodes in which all payments are actually honored, but a sovereign makes a rescheduling offer on less favorable terms than the original debt (Reinhart and Rogoff 2009, citing practices by Moody's and S&P). For a further discussion on such "distressed exchanges," see Section 2.B, above.

B2 to Caa1 (Moody's Global Credit Research 2008). A longer-running saga—that of Argentina's payments to exchange bondholders blocked by court injunction, mentioned above in Section 1.A—was also labeled by some observers, such as the United Nations Conference on Trade and Development, as a "technical" one (UNCTAD 2016).

Repudiation

Repudiation, described in Section 1.C, is rare in modern times.[12] Repudiations most often occur after a regime change, and examples of repudiations include those in the wake of communist revolutions—such as Russia in 1917 or China in 1949.[13] Repudiation is occasionally associated with the concept of "odious debt," which posits that a government is not obligated to pay for those debts incurred by a previous government contrary to the interests of the public.[14] Though the concept has strong moral appeal, it is difficult to define odious debt in a sufficiently limited way to allow its practical application (Reinhart and Rogoff 2009).[15] The doctrine has not gained traction with arbitrators, courts, or credit rating agencies (Gelpern 2007; Blair 2014) and, the earlier example of Ecuador aside,[16] states tend not to assert it explicitly.

Hard vs. soft

Defaults are often categorized as either "hard"/"unilateral" or "soft"/"negotiated," though the exact meaning of these terms often depends on the speaker. The key consideration across the board, however, is whether the debtor is presenting creditors with nonpayment as a fait accompli, or proactively engaging with its creditors on the terms of a default and restructuring. As discussed in Section 1, this is a set of definitions used in the literature that conflates default (an event) with actions taken during the restructuring (a process). However, because this group of definitions is so widely used, it is important to understand it.

[12] In an earlier era, the repudiation of a predecessor government's debt was a common feature in treaties ending armed conflict—e.g., Treaty of Campo Formio of 1797 (France and Austria). Other repudiations followed significant regime changes—e.g., Spain in 1824.

[13] Moody's Sovereign Analytics (2008) and Sturzenegger and Zettelmeyer (2007) provide examples.

[14] For a further examination of this doctrine, see Blair (2014), Gelpern (2007), Buchheit et al. (2007), Jayachandran and Kremer (2006).

[15] To get around the difficult determination of which particular debts should be considered odious, Bolton and Skeel (2011), for example, propose an "odiousness of the regime" approach rather than the traditional "debt-by-debt" approach.

[16] See Feibelman (2010) for an argument that the government's justification for default does not meet the traditional definition for "odious debt" in that it does not show that "the citizens did not obtain meaningful benefits from the underlying transactions."

The literature generally depicts a "hard" or "unilateral" default as a combination of payment default and an aggressive restructuring posture, where the debtor refuses to negotiate with creditors in good faith (Andritzky 2006; Enderlein et al. 2012; Trebesch and Zabel 2017). Such cases may be more closely associated with creditor lawsuits, deep net present value reductions, and capital controls. Debtor–creditor interactions take place against the background of large information asymmetries.

In a "soft" or "negotiated" default, by contrast, the debtor engages proactively with creditors to reach a consensual solution.[17] Generally, a soft default would allow for comprehensive market soundings and informal negotiations with creditors, information sharing, and an offer that could take a menu approach with different options. Of course, this classification is subjective—one man's market sounding is another's take-it-or-leave-it offer. In reality, defaults and restructurings fall somewhere between these two extremes. Examples of defaults often classified as "hard" include the cases of Argentina in 2001 (see Chapter 8) and Russia in 2000 (see Case Study 7.2), while Uruguay's 2003 restructuring provides an example of a "soft" approach (see Case Study 7.2).

Partial vs. full

The literature has also differentiated between "partial" and "full" defaults, typically by considering the amount being defaulted on, though there is significant disagreement over where the boundary lies. Some authors, including Arellano et al. (2019), consider that sovereign debtors only ever partially default, as debtors will always continue to pay some portion of their debt—and often continue to borrow new amounts. Others, such as Eichengreen (1991), consider countries that miss "more than a small fraction of interest payments" to be "heavy" or full defaulters.

A very related concept is that of "selective default," where sovereigns default on some creditor classes while sparing others. For a detailed discussion see Erce (2012) and Schlegl et al. (2017).

Haircut style: rescheduling vs. face value reduction

When default coincides with restructuring, it is often classified according to the extent of ultimate creditor losses. At one end of the spectrum are deep restructurings, where creditors receive new instruments reflecting significant

[17] Where the debtor continues to abide by the payment and other terms of the contract, this would not constitute a legal default. For a further discussion of distressed exchanges, see Section 2.A, above.

net present value (NPV) reductions, or "haircuts." The underlying purpose of such restructurings is to reduce the overall sovereign debt burden, which typically entails principal write-offs and large creditor losses. At the other end of the spectrum are light restructurings, which preserve the principal and most other terms of the original debt contract, but extend its maturity. These operations are functionally identical to rolling over an existing obligation, and are variously called "rescheduling," "refinancing," and most recently, "reprofiling" (these terms are used interchangeably and mean the same thing). Unlike restructurings, reschedulings typically involve small NPV reductions (low haircuts) (Cruces and Trebesch 2013). Both restructurings and reschedulings entail departure from the original debt terms—the difference is of degree.

The most important decision when calculating haircuts is which discount rate to use to calculate the present value of the debt payment streams. The convention in the economic literature, following Sturzenegger and Zettelmeyer (2008), has been to use the "exit yield" for both the old and the new debt, meaning the yield on the newly issued instruments on the first day they are traded. This differs from the approach used in corporate bankruptcies by rating agencies such as Moody's or Standard & Poor's (S&P), which typically take the secondary market price of a bond 30 days after the default to estimate the size of haircuts (or 1 minus the recovery rate). The approach of Sturzenegger and Zettelmeyer (2008) is more comparable to the concept of "ultimate recovery" in corporate finance, also used by Moody's, which focuses on the market price of a bond at the emergence from default (see Meyer at al. 2019 for a discussion and a comparison between the concepts).

While an NPV-based concept of haircuts is a useful metric to compare the depth of different restructurings, it is important to keep in mind that such a computed haircut may not—and generally does not—measure the actual loss (or gain) for a creditor in a debt exchange. The realized loss or gain for a creditor in a trade is based on the price at which they bought the security, compared to the price at which they sold that security or the new security they received in a debt exchange (or its market value in the case of calculating the unrealized loss/gain associated with the trade). Creditors can make large financial gains through a restructuring if they buy the debt at deeply discounted prices in the secondary market before the exchange, and the new securities received in the exchange rally after its completion, perhaps because restructuring terms turn out to be better than what the market had priced, or because the debtor's creditworthiness is perceived to have significantly improved. New research suggests that investors are generally compensated for

the risk of defaults and haircuts. Using long-run data, Meyer et al. (2019) show that investors holding a broad portfolio of risky sovereign bonds achieve a risk–return ratio comparable to that of US equities, despite frequent defaults and major crises.

3. Why Do Defaults Occur?

This section examines the drivers of sovereign default. The question of why and when governments default can also be turned around, by asking: "Why do countries ever repay?," which is arguably *the* fundamental question in the literature on sovereign debt. As emphasized by Bulow and Rogoff (1989a), reputation concerns and purely economic considerations are probably not sufficient to explain why debtors repay. Legal and institutional considerations, in particular enforcement threats, are important to understand the determinants of default and repayment (see also Uribe and Schmitt-Grohe 2017 and Schumacher et al. 2018). The relative importance of enforcement and reputation sanctions in sovereign decisions to repay or default remains at the heart of ongoing debates in economics (see e.g., Tomz 2007; Mitchener and Weidenmier 2010; Aguiar and Amador 2014).

To set the stage, one can think of debt crises as a result of either "mismanagement," meaning bad financial and macroeconomic policymaking at home, and/or of "misfortune," mainly due to external shocks such as a sudden spike in global interest rates, crises in financial center countries, commodity price swings, or natural disasters. In practice, a clean distinction between these two causes is difficult. In particular, it is well-known that countries can implement precautionary macroeconomic and fiscal policies that help to buffer and manage external shocks when they occur. Despite this, it is useful to summarize the findings from the early warning literature on defaults by looking at domestic determinants (Section 3.A) and external determinants (Section 3.B) separately. This distinction can also be applied to the Eurozone debt crisis of 2010–12, which has been characterized by some as a crisis of economic fundamentals and reckless over-borrowing by domestic politicians, while others emphasize the role of cross-border contagion, debt runs by foreign investors, and self-fulfilling default expectations (see Section 3.C). In this context, we will also summarize studies on legal drivers of default in Europe and beyond, in particular the impact of bond clauses and jurisdiction choice for sovereign risk and recovery rates (Section 3.D).

A. Mismanagement: domestic determinants of default

To study the domestic determinants of sovereign default Manasse and Roubini (2009) distinguish between *liquidity* and *solvency*. Simply put, liquidity crises are episodes with rollover problems, meaning difficulties in refinancing short-term debt, while solvency crises are marked by a high debt burden.[18] Their analysis shows that the risk of default due to *illiquidity* is especially high if short-term debt exceeds 130 percent of reserves. Above this threshold, defaults can even occur at moderate levels of debt to GDP.

With a view to *insolvency*, Manasse and Roubini (2009) find that the risk of default is particularly high in case of a high stock of *external* debt (in excess of 50 percent of GDP), while the level of total public debt to GDP is a less useful warning indicator. In line with this, Reinhart et al. (2003) show that a high public debt burden alone is not a good predictor of when and why countries default. Using long-run data they show that some countries are "debt intolerant" and have defaulted at debt/GDP ratios of just 20 percent, while others have tolerated debt stocks above 100 percent of GDP for decades without running into distress. At the same time, the authors show that the *credit history* of a country is a crucial predictor of default. All else equal, advanced economies and emerging market countries that have never defaulted are much less likely to run into debt problems than "serial defaulters," with a high historical default probability.

The risks of external debt are also studied by Catão and Milesi-Ferretti (2014) who show that the ratio of net foreign liabilities (NFL) to GDP is an important predictor of sovereign debt crises, especially if this ratio surpasses 50 percent. The recent Eurozone crises fit into this picture, as much of the debt of countries such as Greece or Portugal was owed to foreign, not domestic creditors. *External dependence* thus appears as a dominant explanation for serial debt problems, not just in Europe, but also in Argentina and many other countries that have relied on foreign debt and defaulted again and again (Reinhart and Trebesch 2016).

Beyond solvency and liquidity indicators, recent years have brought to the fore another domestic trigger of sovereign distress, namely banking crises and *sovereign-financial "doom loops"* (Farhi and Triole 2012). Using 200 years of data, Reinhart and Rogoff (2011) show that domestic banking crises are often followed by a sovereign debt crisis, partly due to the large fiscal costs associated with a financial crash. Negative spillovers from bank balance sheets to

[18] For sovereign debtors, the distinction between illiquidity and insolvency is blurry, as liquidity crises can result in a situation of insolvency, while crises of insolvency are often triggered by refinancing problems. See Section 3.C below and Chapter 4.

the sovereign also played a major role during the Eurozone crisis, as documented by Acharya et al. (2014). Announcements of large financial bailouts between 2010 and 2012 were followed by a strong and immediate increase of sovereign risk measures such as CDS spreads. Similarly, Ang and Longstaff (2013) show that systemic sovereign risk is highly related to financial sector distress, rather than macro fundamentals. Finally, economists have also identified domestic macroeconomic volatility (Catão and Kapur 2006) or domestic political and institutional factors as relevant drivers of sovereign risk (Kohlscheen 2007; van Rickeghem and Weder 2009; Enderlein et al. 2012; Trebesch forthcoming).

B. Misfortune: external determinants of default

External shocks are a main reason why countries default on their debt, especially during systemic debt crises that occur in multiple countries simultaneously (Kaminsky and Vega Garcia 2016). Sturzenegger and Zettelmeyer (2006) study the main sovereign default clusters of the last 200 years and find external factors to be decisive, in particular (i) a worsening of the terms of trade; (ii) a recession in the core countries that acted as providers of capital; (iii) an increase in international borrowing costs, for example, due to tighter monetary policy in creditor countries; and (iv) a crisis in an important country that causes contagion across trade and financial markets. These findings are in line with Reinhart et al. (2016, 2018) who show that a collapse of international capital flows and commodity markets are a powerful predictor of default. Five of the six main waves of external default since 1815 were preceded by such a "double bust," meaning a sudden stop in global capital flows that coincides with a collapse in global commodity prices. Kaminsky and Vega Garcia (2016) further show that terms of trade and export shocks have typically preceded defaults in Latin America, while Hilscher and Nosbusch (2010) show the volatility of terms of trade shocks to be a main driver of sovereign bond spreads. The role of sudden stops in capital flows is further examined by Mendoza (2010), while the link between commodity prices and sovereign risk is studied in a more granular way by recent work of Hamann et al. (2018) and Dominguez et al. (2018).

C. Can debt crises be self-fulfilling?

The idea that debt crises could be self-fulfilling goes back to Calvo (1988) and Cole and Kehoe (2000), among others. They show that the probability

of default largely depends on investor expectations and that there can be multiple equilibria in crisis times. During the Eurozone crisis, this notion has regained new prominence. De Grauwe and Ji (2013), for example, show evidence that, at the peak of the crisis, sovereign spreads were mostly driven by market sentiment and had decoupled from macroeconomic fundamentals or risk indicators such as debt/GDP. Beirne and Fratzscher (2013) also find that a large part of the increase in the level and dispersion of bond spreads during the Eurozone crisis cannot be explained by fundamentals. They distinguish between "pure contagion" or herding panics and "fundamentals contagion," meaning a crisis-induced increase in market sensitivity to fundamentals, which was the main contagion channel during the Eurozone crisis according to their results (see Dell'Ariccia et al. 2006 for a similar result for emerging markets after the Russian crisis). Bocola and Dovis (2016) provide a more theory-driven assessment on the role of fundamentals versus self-fulfilling crisis expectations. They find that rollover risks (or self-fulfilling crisis risk) can explain only a small part of the Italian bond spreads during the crisis, while economic fundamentals play the dominant role.

The overall take away from recent research is that there can indeed be more than one equilibrium in crisis episodes and that excessive debt accumulation and weak fundamentals can therefore "leave sovereign borrowers at the mercy of self-fulfilling increases in interest rates" (Lorenzoni and Werning 2013). Precautionary policies ex-ante can help countries to avoid entering this "crisis zone" in the first place, for example via a fiscal rule (e.g., Conesa and Kehoe 2016; Lorenzoni and Werning 2013), while cross-border bailouts or central bank interventions can prevent or mitigate self-fulfilling dynamics ex-post (e.g., Corsetti and Dedola 2016; Corsetti et al. 2017; Roch and Uhlig 2018). Furthermore, Chamon (2007) suggests relying on state-contingent debt (such as GDP-linked bonds) to reduce the likelihood of self-fulfilling runs.

D. Legal determinants of default: do contract terms matter?

Policy and academic debates about sovereign debt contract reform occasionally imply that contract terms which make debt restructuring more orderly, such as CACs, should also make default easier, more attractive, and therefore more likely. However, studies have failed to find consistent evidence that terms described by market participants as "legal" or "process" terms—capturing all

but the core economic bargain—increase sovereign debt prices at issue (e.g., Becker et al. 2003; Eichengreen and Mody 2004).

The puzzle that CACs and related terms have no (or limited) impact on bond pricing has occasionally been explained as a matter of offsetting effects: default may be more likely, but recovery values are higher if the subsequent restructuring process goes smoothly. However, if creditors find smoother restructuring attractive, the "upside" of process terms should become more salient as the sovereign slides into distress (default probability approaches 100 percent), so that contracts with CACs and similar terms that facilitate orderly restructuring should be priced more favorably (see e.g., Carletti et al. 2018). Instead, studies find growing price penalties for process terms as default draws near (e.g., Carletti et al. 2016; Chamon et al. 2018).

Future studies could help illuminate the relevance of contract terms for the default probabilities and recovery values. Interviews with debtors, creditors, and other market participants suggest that they associate legal or process terms with recovery values, but view their impact on the probability of default as simply too uncertain an issue (Gelpern et al. 2019).

4. The Cost of Default

Sovereign defaults can be costly for governments and investors alike and cause collateral damage to the economy of a defaulting country. This section summarizes these costs.

A. Loss of market access

The theoretical literature typically assumes that defaults and distressed restructurings lead to the exclusion of sovereigns from international capital markets, as well as to an increase in borrowing costs afterwards (see, e.g., Eaton and Gersovitz 1981; Arellano 2008). The empirical results, however, are mixed.

Overall, there is a consensus that defaults do hurt the conditions under which governments can borrow abroad and at home, but there is disagreement around how persistent this effect is. For example, the survey by Panizza et al. (2009) indicates that defaults increase borrowing costs (risk spreads) markedly, but only in the first two years post-default. Similarly, Gelos et al. (2011) document that most defaulters regain access to international markets within just one or two years after a crisis. These findings are in line with

older studies[19] and suggest that investors have short memories. In contrast, more recent work by Cruces and Trebesch (2013) and Catão and Mano (2017) account for the severity of default, measured by the size of haircuts or the length of the default, and find evidence for a more persistent, sizable default premium, of 200 basis points, and a longer exclusion period from international markets.

One channel by which defaults affect market access and borrowing costs are credit rating downgrades. It is well-known that ratings decrease markedly before and after sovereign default events (see, e.g., S&P 2018). Post-default ratings can also remain low for long periods, deterring institutional investors from buying and holding these low-rated bonds. Indeed, defaults and downgrades can result in portfolio relocation effects, also because distressed debt instruments are often excluded from benchmark indices. JP Morgan's EMBI index, for example, drops bonds when they become illiquid and have unreliable pricing, which is often the case in default, especially in protracted defaults. Give the current boom in index investing, these types of index exclusions are likely to be increasingly costly for sovereigns in distress.

B. Collateral damage to the economy

The idea that default may cause "collateral damage" to the economy is nothing new. Cole and Kehoe (1998) develop a model of generalized reputation which suggests that default triggers reputational spillovers that adversely affect not only the sovereign credit market but also other fields of the economy (see also Bulow and Rogoff 1989b). In line with this, a large literature has studied the link between default and various economic outcome variables.

First, sovereign debt crises are accompanied by a significant drop in *economic growth*, as shown in more than a dozen studies. Borensztein and Panizza (2009), Furceri and Zdzienicka (2012), and Kuvshinov and Zimmermann (2016), for example, use cross-country panel data to show that defaults are associated with 2 to 6 percentage points lower growth in the first years of the crisis. There is also a consensus that the fall in output is particularly large when defaults are accompanied by banking crises (see, e.g., De Paoli et al. 2009; Kuvshinov and Zimmermann 2014). Furthermore, recent work has

[19] The influential studies by Lindert and Morton (1989) and Özler (1993) find that the average default penalty is not sizable, and leads to an average increase in spreads of, at most, 50 basis points in years one or two after the crisis. Additional evidence, going back farther in history, is provided by Jorgensen and Sachs (1989).

zoomed in on the aggregate relationship between default and growth. Levy Yeyati and Panizza (2011), for example, use quarterly data to show that, on average, output contractions *precede* defaults and that the recovery starts right after the default. Tomz and Wright (2007) show that the relationship between default and output since 1820 is unexpectedly weak and that countries have also defaulted in "good times." Trebesch and Zabel (2017) show that the output losses are more pronounced in "hard" defaults.

Moreover, the literature has documented a negative correlation between default and (i) trade, (ii) foreign direct investment, and (iii) domestic firms in the defaulting country. Regarding trade, Rose (2005) estimates a gravity panel and shows that, following sovereign debt restructurings, trade falls bilaterally by about 7 percent per year and for more than ten years. Both Martinez and Sandleris (2011) and Mitchener and Weidenmier (2010) show that the observed drop in trade during debt crises is mostly due to a "general" effect rather than a bilateral punishment channel. In a similar setup, Fuentes and Saravia (2010) show that countries that undergo a debt restructuring see their FDI flows drop by up to 2 percent of GDP per year. Recent work by Asonuma et al. (2016) suggests that the type of default influences the size of the decline in exports.

For domestic firms, Hébert and Schreger (2017) use data from Argentina to show that a sovereign default significantly reduces the value of domestic firms on the stock market, especially for exporters and foreign-owned companies. In earlier work, Arteta and Hale (2008) and Das et al. (2012) find that sovereign debt crises are accompanied by a sizable drop in external borrowing by domestic firms. This indicates that corporations in defaulting countries face difficulties in accessing foreign capital markets. Borenzstein and Panizza (2010) and Zymek (2012) also focus on this credit channel and provide evidence that defaults hurt those sectors and exporters most that are dependent on foreign financing. These findings are in line with the theoretical model of Mendoza and Yue (2012) in which defaults increase the cost of borrowing abroad and, thus, the cost of paying for imported inputs. The resulting shift to domestic inputs causes efficiency losses in domestic production and lowers growth.

C. Spillovers to the domestic banking sector

Sovereign default can cause major damage on banks and other systemically relevant institutions, especially if they hold large amounts of government

debt. This type of "top-down" sovereign-financial spillover has played an increasingly important role in recent years, most visibly during the Eurozone crisis. Indeed, there is a consensus that heightened sovereign default risk in economies with a large financial sector can result in an aggregate credit shortage, less investment and possibly a banking crisis and an output decline (e.g., Acharya et al. 2014; Perez 2015; Bocola 2016; Engler and Grosse Steffen 2016, Sosa-Padilla, forthcoming).

A widely cited paper on the link between sovereign default and banks is Gennaioli et al. (2014), who find that sovereign defaults are followed by large drops in private credit and that this post-default credit crunch is stronger for countries in which banks hold more government debt. In follow-up work, the same authors use finer-grained data and again find a strong negative correlation between a bank's holdings of government bonds and its lending during sovereign defaults (Gennaioli et al. 2018). Acharya et al. (2018) and Bofondi et al. (2017) reach a similar result when linking data on a bank's holdings of sovereign debt and that bank's lending activity.

D. Creditor lawsuits: the legal costs of default

Other important concerns for policymakers in the context of default are legal risks (threat of litigation) and the costs arising from the so called "holdout problem" (see Chapter 8; Panizza et al. 2009; Buchheit et al. 2013). Overall, the evidence shows that sovereign immunity has eroded since the 1970s, strengthening the hands of creditors and raising the legal cost of default for debtors, with implications for government willingness to repay.

The Argentine debt crisis after 2001 is the best-known example of how large the legal costs of default can become. Dozens of hedge funds filed suit against Argentina in New York, litigated for full repayment, and repeatedly attempted to seize Argentine assets abroad. Fifteen years later, those holdout creditors achieved a major victory in court, which ultimately forced the Argentine government into a settlement of more than $10 billion—a multiple of the debt's original face value (Hébert and Schreger 2017).

Argentina is not an exception but is part of a general trend, as shown by Schumacher et al. (2018). Building on a new dataset on sovereign debt lawsuits, these authors document that the risk of litigation in the context of a sovereign default has greatly increased since the 1980s.[20] Furthermore, they

[20] The risk of litigation is particularly high for sovereigns imposing a high haircut on large amounts of debt (Schumacher et al. 2015).

show that legal disputes can disrupt government access to international capital markets, as foreign courts impose a financial embargo on defaulting sovereigns. Legal risks are therefore one possible channel explaining why governments lose market access.

In recent years, the risk of creditor holdouts and litigation has only continued to increase. Schumacher et al. (2018) document that, unlike in the 1990s or early 2000s, governments in distress now frequently point to legal risks when explaining their policy choices, and the same is true for rating agencies justifying up- or downgrades. In line with this, Buchheit et al. (2013) argue that the fear of a protracted legal disputes with creditors, as in Argentina, was one of the reasons many Eurozone governments decided to avoid a default or debt restructuring. The only Eurozone default occurred in Greece, but even there legal risks played an important role. Confronted with legal risks, the Greek government decided to repay holdouts on foreign-law bonds in full and on time, allowing them to escape the haircut imposed on all other creditors. The resulting transfers amounted to more than 2 percent of Greek GDP (Zettelmeyer et al. 2013).

5. Reducing the Incidence and Costs of Default

This section examines the literature on experience with the incidence and cost of default, although it necessarily discusses the process of restructuring as well. The section is organized as follows. Section 5.A documents the "too little, too late" problem: whether or not a country defaults, restructurings are often delayed and when they do take place they often don't entail a deep enough restructuring to definitively restore sustainability. Hence they end up being more costly. Section 5.B reviews some possible factors behind such outcomes, discussing both the incentives of debtors as well as official creditors. Finally, Section 5.C briefly discusses what can be done and has been done to make default and restructurings less likely and to reduce their associated costs.

A. Too little, too late: timing and depth of restructuring

Sovereigns, as well as other key stakeholders in a restructuring process such as official creditors, have a tendency to delay defaults and necessary restructurings (the "too late" problem). According to IMF (2014), about 80 percent of the countries that experienced a restructuring had a sustained high debt

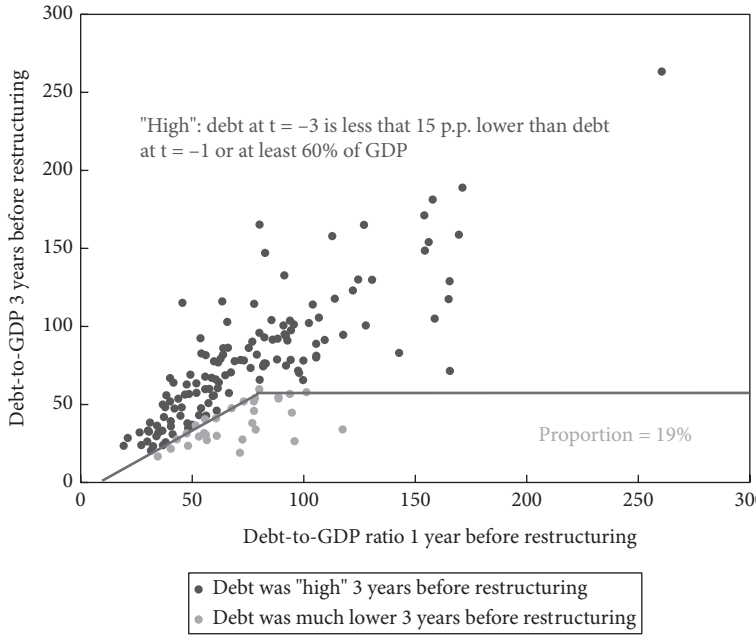

Figure 7.2 Delayed restructurings
Source: IMF (2014).

level three years before the date of the restructuring (Figure 7.2). Delayed restructurings are costly, first and foremost to debtors but also to creditors. House et al. (preliminary, 2017) find that the longer it takes to announce a restructuring, the bigger the cumulative output losses, the larger the haircuts for creditors, and the more costly the subsequent borrowing for the sovereign. For debtors, a situation of debt overhang depresses investment and growth and creates a sense of financial uncertainty that can raise the eventual magnitude of resolving the debt problem. For countries experiencing debt distress and which are considering approaching the official sector for assistance, creditors have an incentive to lend on shorter maturities recognizing that they have a higher chance of being bailed out. This in turn distorts the incentives of the country to favor short-term borrowing, further worsening its debt profile. For creditors, delayed restructurings, particularly when some private creditors are paid out in the interim, imply that those that are left, or who lent on longer maturities, will have to absorb a greater loss. Case Study 7.3 provides the example of Greece in 2012. This also applies to official creditors who may themselves initially want to delay addressing a debt problem through a restructuring but end up increasing risks to their own prospects of being

Case Study 7.3 Greece 2012—delayed restructuring

A prominent example of a delayed restructuring, exacerbated by external factors, is Greece. As explained in detail by the IMF's Independent Evaluation Office (2016), when Greece approached the IMF in May 2010, its debt situation did not meet the bar required by the IMF to lend it large amounts. Instead of withholding IMF financing until Greece undertook a debt operation that would bail in private creditors—the normal requirement under IMF policies—it was decided in the context of approving the loan to Greece to change the IMF's policies to allow it to bail out private creditors with official resources. The key motivation underlying this decision was the fear that a bail in of private creditors—primarily European banks—would raise the costs of resolving the crisis through contagion to other high-debt euro area sovereigns and banks given the significant systemic spillover risks. Besides, there was a chance that Greece might be able to pull itself out of its difficulties and regain market access. In the event, contagion worsened after the bail out, Greece's debt profile became more rigid due to a higher share of official debt, and the remaining private creditors received a deep haircut in 2012—deeper than would have been necessary if other private creditors had not been paid out in the interim.

repaid, especially if their resources are used to pay out private creditors during a prolonged period of debt distress.

When restructurings have taken place, they have often failed to restore debt sustainability and market access, leading to repeated restructurings and dependence on official financing (the "too little" problem). The literature finds that more comprehensive initial restructurings lessen the likelihood of repeat restructurings—relatively low haircuts often presage serial restructurings (Reinhart et al. 2003; Schroeder 2015; Mariscal et al. 2015; Asonuma 2016; Ghosal et al. 2016). A famous example from earlier times is Poland. It went through six restructurings with private creditors and four with the Paris Club between 1981 and 1990, mostly consisting of rescheduling of principal and interest. The debt problem was not fully resolved until the Paris Club granted 50 percent debt forgiveness in 1991 and, after lengthy negotiations, private banks agreed to a 45 percent debt reduction in 1994. According to IMF (2014), from 1980 to 2012, of the forty-four countries that restructured their debt, 86 percent had more than one restructuring. This pattern emerged in

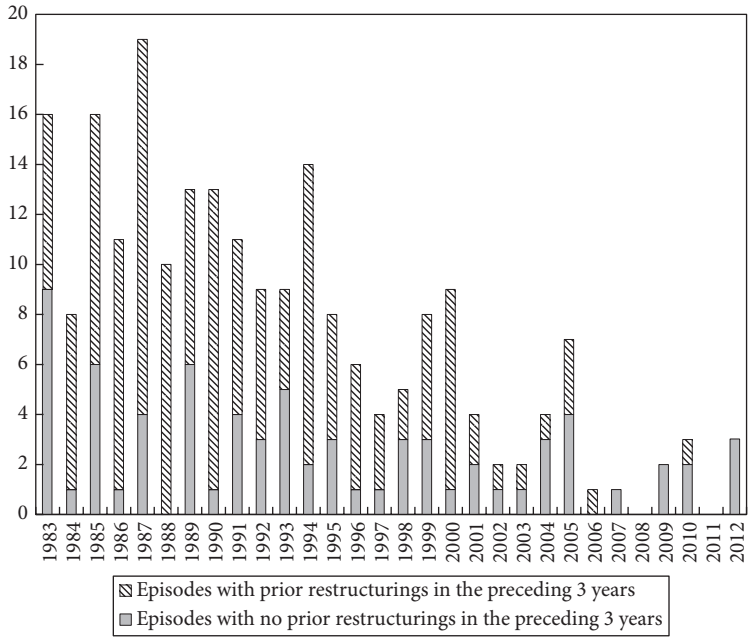

Figure 7.3 Repeat restructurings

Source: IMF (2014); Reinhart and Rogoff (2009); Das et al. (2012); and Cruces and Trebesch (2013).

restructurings with both private and official creditors; on average each country had over five restructurings, of which about half were with private foreign creditors (Figure 7.3). Repeat restructurings suggest that a one-time restructuring was often not enough to solve the debt problem. On average, two-thirds of all restructurings with private foreign creditors did not successfully establish sustainability and led to repeat restructurings.

B. Causes of delayed and inadequate restructurings

Why are restructurings delayed and often insufficient to definitively restore sustainability? The latter part of this question may be easier to answer. Inadequately-sized restructurings may be linked to the desire to avoid the costs associated with large debt reductions. The available literature finds that significant debt reductions are associated with higher economic costs and lengthier market exclusion (Cruces and Trebesch 2013; Trebesch and Zabel 2017). However, while it is intuitive that a deeper restructuring may cause greater costs and spillovers, it is not clear why repeated restructurings are considered a better outcome than a "one-and-done" approach. Indeed,

Reinhart and Trebesch (2016) find that countries in a situation of chronic debt overhang tend to grow again only after significant debt relief agreements involving face value reductions, such as the Brady deals of the 1990s or the cancelation of war debts after 1932. Similarly, Mariscal et al. (2015) find that shallower debt treatments, which have a lower cost individually, may not solve the underlying debt problems, noting that over 40 percent of such treatments are followed by another within six years.

With regard to bilateral official creditors, the incidence of many repeat restructurings may reflect the political difficulties associated with giving a principal reduction. Repeated flow treatments have been a common experience of the Paris Club as in Poland, as noted earlier. Moreover, the countries that have gotten significant stock reductions have all been politically significant cases: Poland (in 1991 to woo it away from the Soviet Union after the break up), Egypt (to reward it for its support to the western countries during the 1991 Persian Gulf War), and Iraq (after the Second Gulf War).

The reasons why restructurings are initiated with delay may be more complex. On the one hand, an inclination to delay is no surprise. Debtor governments fear the economic, financial, and political fallout of a default and restructuring, particularly if the domestic financial sector hold a significant amount of public debt (e.g., Jamaica). Private creditors will also naturally wish to avoid a debt restructuring if at all possible and will therefore press for a bailout by the official sector. Finally, official creditors may also have incentives to delay a restructuring or default out of concerns that this would reduce incentives for the debtor country to adjust, force banks located in official lenders' countries to recognize losses, and trigger market turmoil affecting similarly situated countries, or to preserve flexibility for the future.

However, when a debt restructuring is the only option to deal with a distressed situation, it is not clear how the debtor country or the official sector help themselves by delaying the inevitable.

C. Reducing the costs of restructurings

It would be naïve to suppose that countries will take heed of the foregoing discussion on the costs of default and always abstain from policies that could put them in a situation of debt distress. It would be similarly naïve to suppose that private creditors will always work with the debtor to find ways for timely and adequately sized restructurings that would be less costly. However, very often, a decision to default or restructure comes about in the context of a lending

program from the official sector, most typically from the IMF. Therefore, the international financial policy architecture and particularly the policy framework of the IMF that governs when and how much it can lend may affect the incidence and cost of default. This section, therefore, examines that framework.

One of the key changes brought about by the IMF in 2016 was the change to its policy for giving large loans (the "Exceptional Access Policy"). Prior to changes made in the context of lending to Greece 2010 (see Case Study 7.3), the IMF's exceptional access policy required that if the IMF could not say with "high probability" that a crisis-struck country's debt position was sustainable, the IMF could not lend absent a debt restructuring sufficiently deep to satisfy this condition. This approach had merit for cases where debt was considered clearly unsustainable. However, it was too rigid for cases where debt was considered sustainable but risks around the debt outlook did not allow this assessment to be made with high probability (i.e., sustainability was in the "gray zone"). In such cases, it could be sufficient to give a chance to a lighter restructuring—a reprofiling or an extension of maturities—which would improve debt sustainability and help the country overcome the crisis without incurring the costs of a deeper restructuring. The key change in the Exceptional Access Policy in 2016 was to introduce such flexibility into the IMF's lending framework and better tailor lending options to the country's debt situation as illustrated by Figure 7.4. In line with the "too little" problem

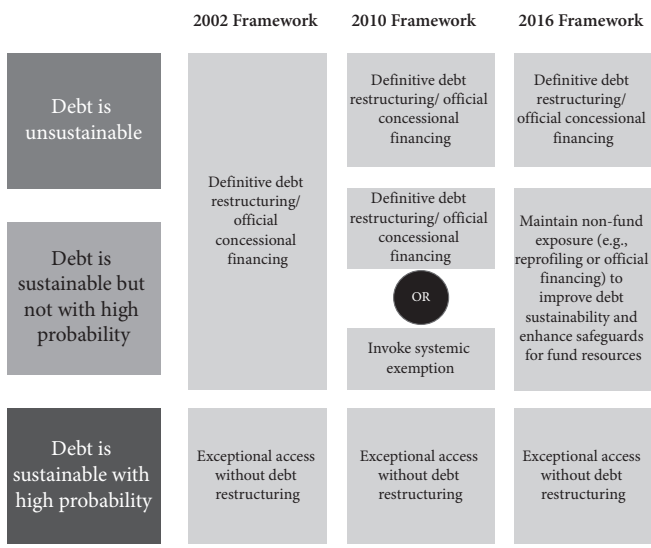

Figure 7.4 Changes to the IMF's lending framework
Source: IMF (2015a)

discussed above, however, the IMF did note that repeat reprofilings should be avoided—if an initial reprofiling failed to dispel concerns, a deeper debt reduction would be needed.

The introduction of such flexibility into the lending framework also allowed the Fund to do away with the "Systemic Exemption." The Exemption was introduced in 2011 in the context of lending to Greece due to contagion-related concerns and allowed the IMF to provide financing in exceptional levels after 2012 if there was "a high risk of international systemic spillovers" (see Case Study 7.3). In hindsight, the experience with the use of the systemic exemption revealed that while there may have been important considerations behind its introduction, it did not help address the underlying problem. By severing the link between underlying debtor risk and yields, the Exemption encouraged moral hazard and over-borrowing ex-ante, and exacerbated market uncertainty in periods of sovereign stress, as traders bet on whether the Exemption would be activated, rather than focusing on underlying sustainability fundamentals. The exemption also reduced safeguards for use of IMF resources, since if a debt restructuring was eventually needed, a smaller pool of private creditors would be available to absorb losses.

Annex 1 Examples of events of default—payment default, covenant default, and cross-default in foreign-law tradable sovereign debt securities

Italy 2013
(New York Law, Fiscal Agency)

Default; acceleration of maturity

Each of the following is an event of default under any series of debt securities:

- We default in any payment of principal, premium or interest on any debt securities of that series and the default continues for a period of more than 30 days after the due date.
- We fail to perform or observe any other obligation under any debt securities of that series and the default continues for a period of 60 days following written notice to us of the default by any holder.
- Any other present or future Public External Indebtedness in an amount equal to or exceeding US$50 million (or its equivalent) becomes due and payable prior to its stated maturity by reason of default in payment of principal thereof of premium, if any, or interest thereon.

- Any other Public External Indebtedness in an amount equal to or exceeding US$50 million (or its equivalent) is not paid at its maturity as extended by any applicable grace period. […]

Kazakhstan 2014
(English Law, Fiscal Agency)
[…] **Events of Default**
The Fiscal Agent shall upon receipt of written requests from the holders of not less than 25 percent in aggregate outstanding principal amount of the Notes or if so directed by an Extraordinary Resolution shall, give notice to the Issuer that the Notes are and they shall immediately become due and repayable at their principal amount together with accrued interest if any of the following events (each, an "Event of Default") occurs and is continuing:

(a) Non-payment: the Issuer is in default with respect to the payment of interest or additional amounts on any of the Notes and such default continues for a period of 30 days; or
(b) Breach of other Obligations: the Issuer is in default in the performance, or is otherwise in breach, of any covenant, obligation, undertaking or other agreement under the Notes, other than a default or breach elsewhere specifically dealt with in this Condition 13 and such default or breach is not remedied within 60 days after notice thereof has been given to the Issuer at the Specified Office of the Fiscal Agent by any holder of Notes; or
(c) Cross Default: (a) any other Public External Indebtedness of the Issuer (i) becomes due and payable prior to the due date for payment thereof by reason of default by the Issuer, or (ii) is not repaid at maturity as extended by the period of grace, if any, applicable thereto, or (b) any Guarantee given by the Issuer in respect of Public External Indebtedness of any other Person is not honoured when due and called upon; provided that the aggregate amount of the relevant Public External Indebtedness or liability under such Guarantee in respect of which one or more of the events mentioned in this Condition 13(c) shall have occurred equals or exceeds U.S.$65,000,000 or its equivalent in other currencies; […]

Mexico 2014
(New York Law, Trust Indenture)

Default and acceleration of maturity

Each of the following is an event of default under any series of debt securities:

- Mexico fails to pay any principal, premium, if any, or interest on any debt security of that series within 30 days after payment is due;
- Mexico fails to perform any other obligation under the debt securities of that series and does not cure that failure within 30 days after Mexico receives written notice

from the trustee or holders of at least 25 percent in aggregate principal amount of the outstanding debt securities requiring Mexico to remedy the failure;

- Mexico's creditors accelerate an aggregate principal amount of more than U.S. $10,000,000 (or its equivalent in any other currency) of Mexico's public external indebtedness because of an event of default resulting from Mexico's failure to pay principal or interest on that public external indebtedness when due;
- Mexico fails to make any payment on any of its public external indebtedness in an aggregate principal amount of more than U.S. $10,000,000 (or its equivalent in any other currency) when due and does not cure that failure within 30 days after Mexico receives written notice from the trustee or holders of at least 25 percent in aggregate principal amount of the outstanding debt securities requiring Mexico to remedy the failure; […]

Ghana 2014
(English Law, Fiscal Agency)
[…] *Events of Default*
If any of the following events ("Events of Default") shall have occurred and be continuing:

(a) *Non-payment*
  (i) the Issuer fails to pay any principal on any of the Notes when due and payable and such failure continues for a period of 15 days; or
  (ii) the Issuer fails to pay any interest on any of the Notes or any amount due under Condition 8 (Taxation) when due and payable, and such failure continues for a period of 30 days; or
(b) *Breach of other obligations*
  The Issuer does not perform or comply with any one or more of its other obligations under the Notes, which default is incapable of remedy or is not remedied within 45 days following the service by any Noteholder on the Issuer of notice requiring the same to be remedied; or
(c) *Cross-default*
  (i) the acceleration of the maturity (other than by optional or mandatory prepayment or redemption) of any External Indebtedness of the Issuer; or
  (ii) any default in the payment of principal of any External Indebtedness of the Issuer shall occur when and as the same shall become due and payable if such default shall continue beyond the initial grace period, if any, applicable thereto; or
  (iii) any default in the payment when due and called upon (after the expiry of any applicable grace period) of any Guarantee of the Issuer in respect of any External Indebtedness of any other person,

provided that the aggregate amount of the relevant External Indebtedness in respect of which one or more of the events mentioned in this paragraph (c) have occurred equals or exceeds US$25,000,000 or its equivalent; […]

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